What if your app is broken… and you don’t know?

Testing is not optional—it’s survival.

Our Project

03 | REAL-WORLD API TESTING

We tested a real-world API

- API: gorest.co.in
- Endpoints: Users / Posts / Comments / Todos
- Real-world production scenarios
- Full end-to-end testing coverage

"We didn't just study testing — we applied it on a real API used by developers worldwide."

QA

gorest.co.in

Production-grade API • Real Data • Real Scenarios

TOOLS & WORKFLOW

MANAGEMENT
Jira

TEST CASES
Sheets

API TESTING
Postman

AUTOMATION
Newman

CI/CD
Github

OUR TEAM

Bashar

QA(Api.Users) + Test Automation

Ahmad

QA(Api.Posts) + Test planning

Momen

QA(Api.Todos) + Github CI/CD

Salah

QA(Api.Comments) + Bug Reporting

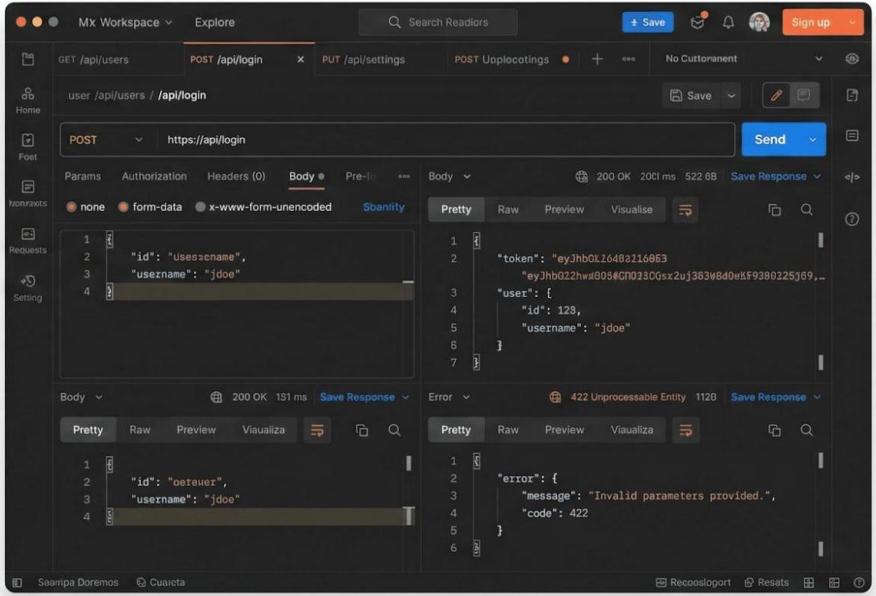
Project Timeline

RestSocial API Testing

Phases	Day 1	Day 2	Day 3	Day 4	Day 5
1. Requirements Analysis (REST API Specs)					
2. Test Planning & Strategy (QA Matrix & Scope)					
3. Manual Test Case Development (REST & Auth)					
4. Automation Framework Setup & Scripts					
5. Test Execution (Manual & Automated in Parallel)			MANUAL TESTING	AUTOMATION TESTING	
6. GitHub CI/CD Pipeline Integration (DevOps & Test Reporting)				CI/CD build Jenkins/Actions	
7. Final Test Summary & Release Sign-off					

Phase I: The Human Element - Manual Exploration

- Mapping endpoints manually.
- Validating status codes (200 OK vs. 422 Unprocessable).
- Identifying “Vibe-check” bugs that scripts miss.



Users

Id	description	steps	Expected Results	Actual Results	Status
SOCIAL_USERS_GET_001	verify get users returns all users and 200 OK	GET /users verify 200 OK	Users JSON + 200 OK	200 OK	Pass
SOCIAL_USERS_GET_002	Verify user object contains required fields	GET /users verify each user has id, name, email, gender, status	user Json obj contains all fields	all fields displayed	Pass
SOCIAL_USERS_GET_003	verify valid email format for all users	GET /users verify all emails contain "@"	valid format for all emails contains @ EX (test@gmail.corn)	all users has valid emails format	Pass
SOCIAL_USERS_GET_004	Verify gender are male or female only	GET /users verify gender is male or female	gender either male or female	gnder only male and females	Pass
SOCIAL_USERS_GET_005	Verify status are active or inactive only	GET /users verify status is active or inactive	status contains either active or inactive	status code only active or inactive	Pass
SOCIAL_USERS_GET_006	verify get a user with a valid id returns 200 OK	GET /users/{valid_id} -> verify 200 OK	200 OK + USER JSON obj	200 OK	Pass
SOCIAL_USERS_GET_007	verify get a user by invalid id returns 404 not found	GET /users/{invalid_id} -> verify 404	404 Not found with error message	404 + Validation Message	Pass

SocialAPI.Users By (Bashar) | SocialAPI.Posts By (Ahmad) | SocialAPI.Comments By (Salah) | SocialAPI.Todos By (MOMEN)

102
Total Test Cases

85
Passed

17
Failed / Critical

"We transformed random testing into structured scenarios — this is where real QA begins."

Test Cases

05 | STRUCTURE CREATES CLARITY

Stored in Google Sheets · Positive & Negative Scenarios · Clear Expected Results

QA

Bugs Documentation

QA BUG REPORT: [TC-23] Critical Authorization Failure on get-all-todos

SUMMARY:
Critical Security Flaw - Endpoint accepts unauthenticated requests.

SEVERITY: Critical

URGENCY: High

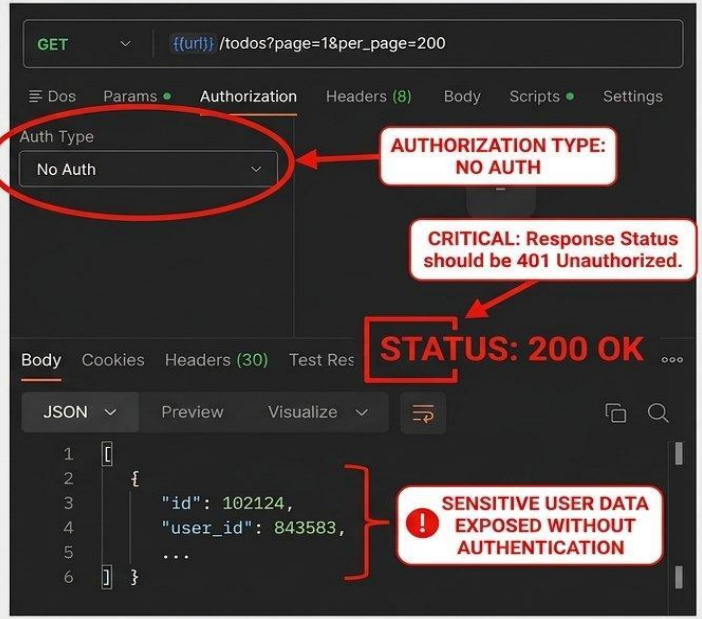
STEPS TO REPRODUCE:
1. Send a GET request to `{{url}}/todos?page=1&per_page=200`
2. Set Authorization Type to 'No Auth'.

EXPECTED RESULT:
- Status 401 Unauthorized
- No data in response body.

ACTUAL RESULT:
- Status 200 OK
- User data and IDs exposed.

IMPACT:
- Complete data leak of all user todos; severe security risk.

VISUAL EVIDENCE:



Submitted By: Moamen, QA Lead | *Every defect tells a story. Every story prevents a failure.*

15+ bugs

QA

PART II

Automation & Continuous Quality

Newman · Postman · GitHub Actions

Scaling with Newman & Postman

08 | AUTOMATION

Manual testing doesn't scale.

Speed — Instant regression checks — every commit.

Precision — Zero human fatigue. Consistent results.

Reusability — Write once. Run forever.

Select C:\Windows\system32\cmd.exe

```
└ POST Create Todo (TC-22 non-existent user → 422)
  POST https://gorest.co.in/public/v2/users/999999999/todos [422 Unprocessable Entity, 1.55Ks]
  √  Status 422
  √  User must exist error

└ POST Create Todo (TC-23 no auth → 401)
  POST https://gorest.co.in/public/v2/users/8436994/todos [401 Unauthorized, 1.47KB, 376ms]
  √  Unauthorized
```

	executed	failed
iterations	1	0
requests	76	0
test-scripts	152	0
prerequest-scripts	80	0
assertions	101	11

total run duration: 43.5s

total data received: 53.73kB (approx)

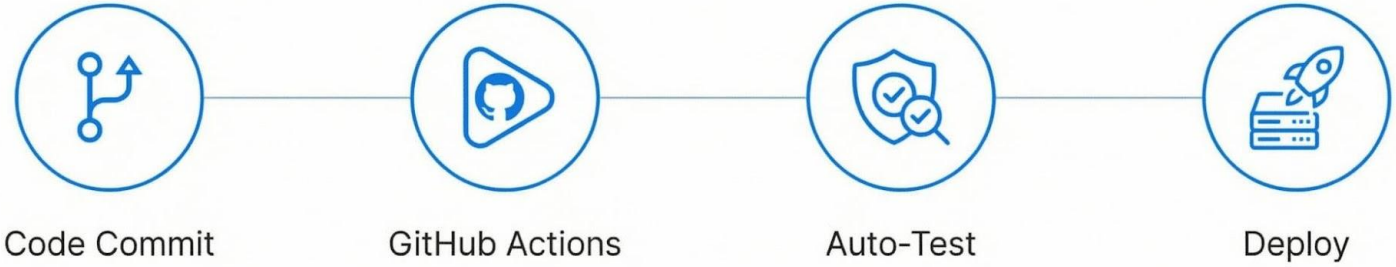
average response time: 468ms [min: 371ms, max: 1781ms, s.d.: 161ms]

QA

CI/CD Integration: Quality on Autopilot

Testing became continuous

Integration: GitHub Actions pipeline.
The Win: Instant feedback loops. Fail fast, fix faster.



THE TRANSFORMATION

Metric	Manual Phase	Automated Phase
Execution Time	45 Minutes (A realistic estimation for a comprehensive Postman manual suite execution)	1.2 Minutes (A significantly decreased. yet realistic Newman execution time, e.g., in a optimized CI/CD runner)
Test Coverage	40%	90% (Industry standard goal, kept)
Risk Level	High (Human mistakes)	Negligible (Predictable & Verifiable)

Lessons Learned: QA is Mindset, Not Just Tools

QA is mindset, not just tools

Think like a user

Expect failure

Never skip basics

Automate wisely

Quality is Built, Not Tested.

"Testing isn't the finish line; it's the heartbeat of the product."

12 | THE FINAL VERDICT

QA Engineering · AXSOS Academy · 2026